

Cloudflare Radar 2024 Year in Review

Internet patterns and trends observed through Cloudflare's global network during 2024

Data-driven analysis of global Internet traffic, adoption trends,
connectivity disruptions, and security threats

Prepared for: Technology Executives & Network Infrastructure Leaders

17% Global Traffic Growth and Starlink's 3.3x Surge

Continuing the upward trend driven by increasing reliance on the Internet for communication, commerce, entertainment, and transportation

+17%

YoY Global Traffic Growth

3.3x

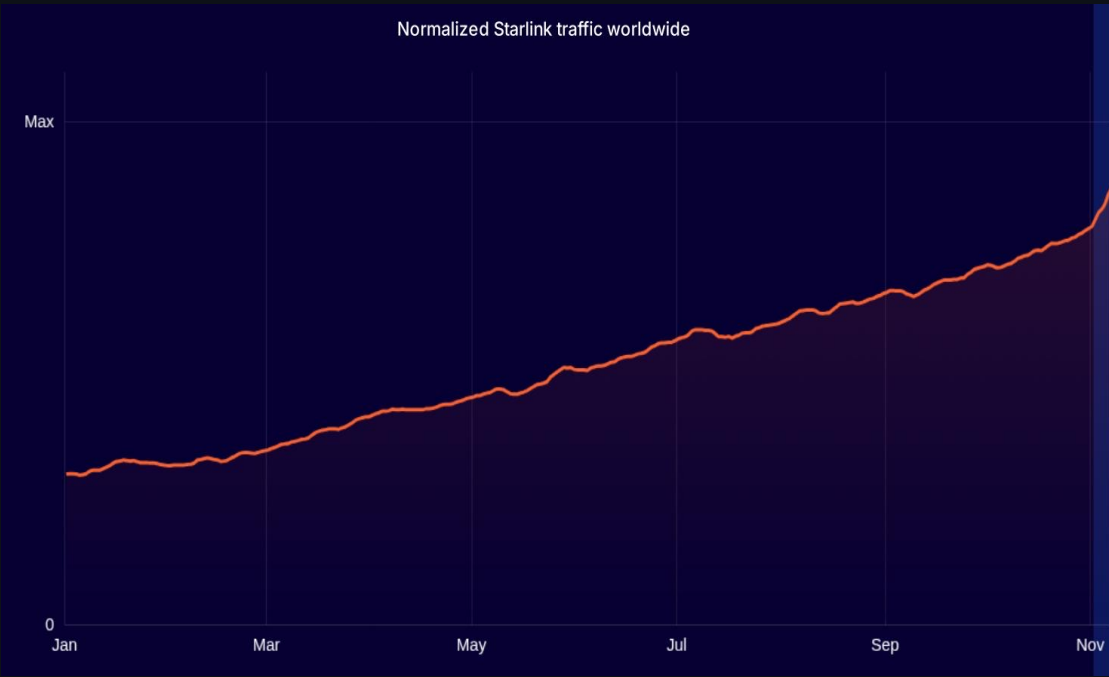
Starlink Traffic Surge

41%

Mobile Device Traffic

28%

IPv6 Adoption Rate



KEY HIGHLIGHTS

- Global Internet traffic grew 17% in 2024, with the trend line starting mid-January to normalize for post-holiday activity.
- SpaceX Starlink traffic surged 3.3x, extending satellite Internet connectivity to previously underserved locations worldwide.
- 41% of global traffic came from mobile devices, reflecting the continued importance of mobile connectivity.
- IPv6 adoption reached 28% of dual-stack traffic. Spain recorded the highest download speed among all countries.

Google Leads Top 10 Services as Generative AI Rises

Ranking based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver | Generative AI continued its rapid rise throughout 2024

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
⚡ Generative AI continued its meteoric rise throughout 2024, while the Metaverse receded into the background	

TOP 10 MOST POPULAR INTERNET SERVICES

#1	Google	Search, Cloud, Workspace
#2	Facebook	Social Media
#3	Apple	Ecosystem & Services
#4	TikTok	Short-form Video
#5	AWS	Cloud Infrastructure
#6	Microsoft	Productivity & Cloud
#7	Instagram	Photo & Video Sharing
#8	YouTube	Video Streaming
#9	Amazon	E-Commerce
#10	WhatsApp	Messaging

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Key technology adoption shifts in protocol support, browser usage, and developer ecosystem during 2024



2% → 13%

Post-Quantum TLS 1.3

Chrome 124

Default PQ Enablement

POST-QUANTUM ADOPTION TIMELINE

- Early 2024: ~2% of TLS 1.3 traffic used post-quantum encryption
- April 17: Chrome 124 enabled post-quantum key agreement by default
- Rapid climb to approximately 12-13% of TLS 1.3 traffic
- Mozilla Firefox rolling out post-quantum by default
- Apple Safari began initial testing

HTTP/3 ADOPTION

21%

of traffic uses HTTP/3, running on QUIC for faster connections, packet-loss mitigation, and encryption by default

Source: Cloudflare Radar 2024 Year in Review | radar.cloudflare.com

TOP API CLIENT LANGUAGES

Go	12%
Node.js	10%
Python	9.6%
Java	7.4%
.NET	3.6%

SEARCH ENGINE SHARE

Google	89%
Yandex	3.1%
Baidu	2.7%
Bing	2.6%
DuckDuckGo	0.95%

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Internet disruptions tracked by the Cloudflare Radar Outage Center using Cloudflare traffic data

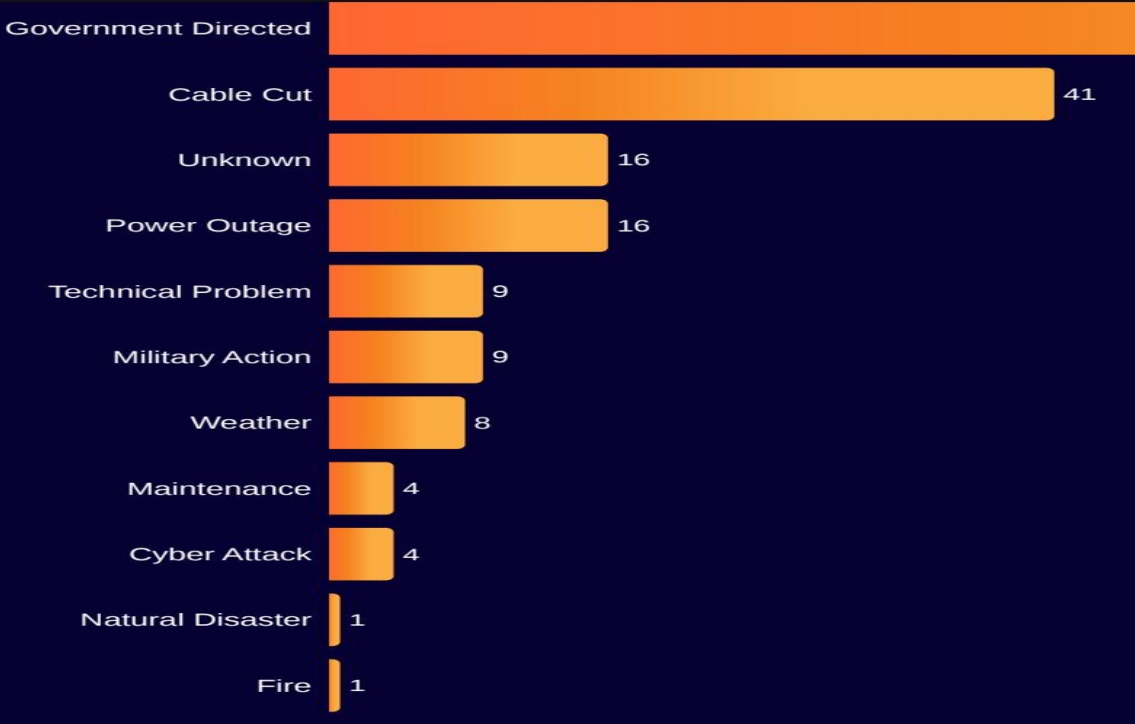
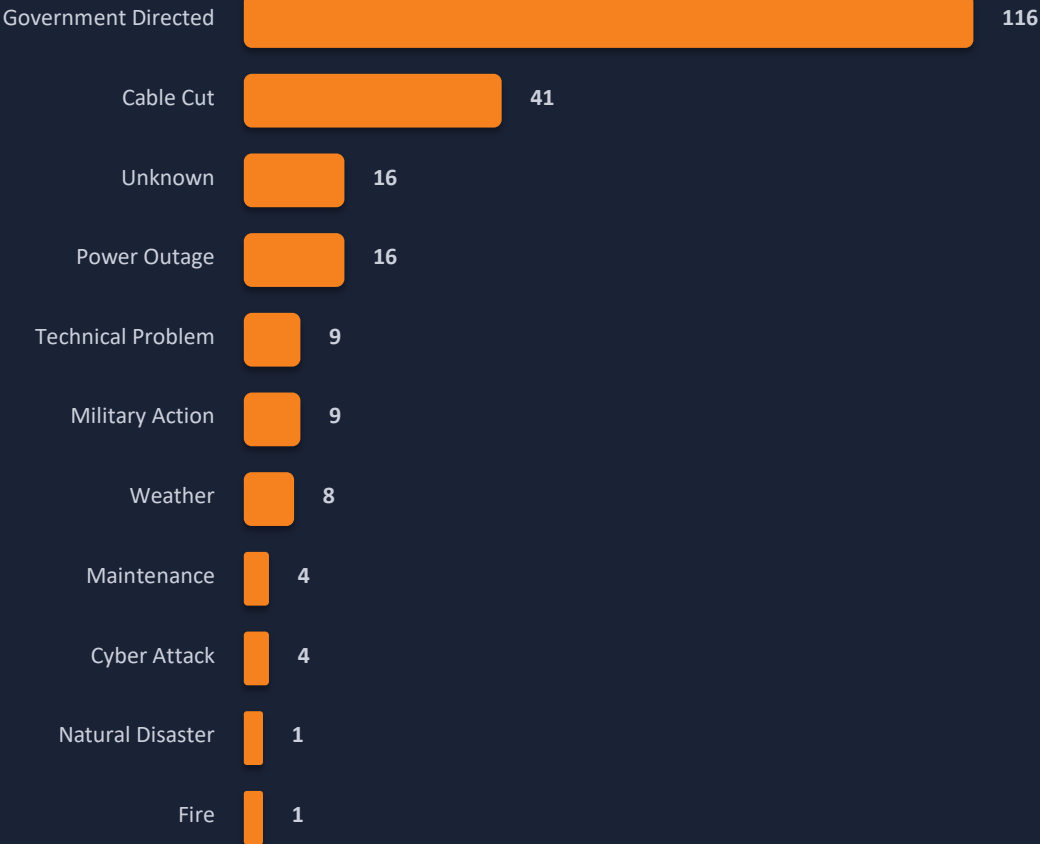
225

Major Internet Disruptions

116+

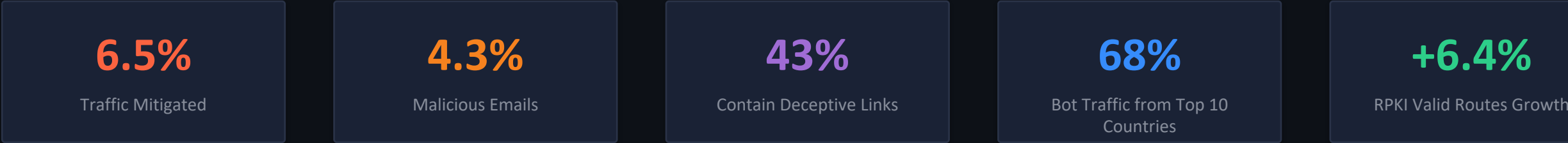
Government-Directed Shutdowns

DISRUPTION CAUSES BREAKDOWN



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Security threat landscape observed across Cloudflare's global network during 2024



SECURITY THREAT HIGHLIGHTS

- Gambling & Games was the most attacked industry sector
- Log4j (CVE-2021-44228) continues to be actively targeted 2+ years after disclosure, underscoring persistent vulnerability exploitation
- 68% of global bot traffic originated from just the top 10 countries, indicating high geographic concentration
- 6.5% of all traffic processed by Cloudflare was mitigated, indicating the massive scale of ongoing threats
- RPKI valid routes increased 6.4%, improving Internet routing security

EMAIL SECURITY LANDSCAPE

- 4.3%**
of emails processed were identified as malicious
- 43%**
of malicious emails contained deceptive links designed to trick recipients
- .bar**
top-level domain originated the largest share of malicious and spam email

Five Takeaways Shaping the 2025 Internet Outlook

Strategic insights from 2024 to inform planning and investment decisions for network and security leaders

01

Traffic Growth Trajectory

Global Internet traffic grew 17% YoY with Starlink surging 3.3x. Satellite and mobile connectivity will continue driving traffic growth, requiring infrastructure investment in capacity and edge computing.

02

Post-Quantum Encryption Momentum

Post-quantum encryption leapt from 2% to 13% of TLS 1.3 traffic after Chrome 124's default enablement. Organizations must begin planning their own post-quantum migration strategies as browser adoption accelerates.

03

Connectivity Fragility from Government Actions

Government-directed shutdowns led 225 major disruptions (116+ incidents). Enterprises operating in affected regions need redundant connectivity paths and resilient architecture to maintain service availability.

04

Persistent & Evolving Security Threats

6.5% of traffic mitigated, Log4j still actively exploited 2+ years post-disclosure, and 68% of bot traffic from top 10 countries. Defense-in-depth strategies and rapid patching remain essential.

05

Evolving Developer Ecosystem

Go became the top API client language (~12% of automated traffic). HTTP/3 reached 21% adoption. Technology leaders should align their stack investments with these shifting developer and protocol preferences.